GEOMETRY · UNIT 1: FOUNDATIONS OF GEOMETRY

Treasure Map Navigator

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students follow clues on a coordinate treasure map using the distance and midpoint formulas, segment and angle addition, and angle pairs, then design a treasure map with clues of their own for classmates to solve.

Students apply the foundational tools of geometry in a puzzle setting. Each clue needs one tool, and the design task asks students to reverse the process and write clues that lead to a chosen point.

Standards Addressed

- G-CO.1 Precise definitions of angle, line segment, and distance
- G-GPE.4 Use coordinates to prove simple geometric theorems
- G-GPE.6 Find the point on a segment that partitions it in a given ratio (Honors)
- G-GPE.7 Compute perimeters with coordinates

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the map: where would you dig? Part 1 distance and midpoint clues.
Day 2	Part 2 segment and angle clues.
Day 3	Part 3 design a map; swap and solve.

Materials

- Student handout and map
- Rulers and protractors
- Graph paper for the student map
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Coordinates given in a table; distance by the Pythagorean Theorem with a grid; one-step algebra in segment and angle addition.
CP1	Distance and midpoint formulas, finding an endpoint from a midpoint, perimeter, and angle bisectors.
Honors	Partitioning a segment in a ratio, the perpendicular bisector as an equidistant set, and a design with an equidistance clue.

Watch For (Common Misconceptions)

- Subtracting coordinates in different orders for x and y.
- Averaging only the x-coordinates for a midpoint.
- Assuming a diagram labeled 'not to scale' can be measured.

Extension Ideas

- Use a real school map with a coordinate overlay.
- Add a compass-bearing clue.
- Make a digital map in Desmos.

Answer Key – CP2

Part 1: Distance & Midpoint Clues

#	Task	Answer
1	Clue 1: How far is it from the Dock to the Cave? (They are on the same horizontal line.)	12 blocks
2	Clue 2: How far is it from the Dock to the Palm? Draw a right triangle on the grid and use the Pythagorean Theorem.	$\sqrt{(8^2 + 6^2)} = \sqrt{100} = 10$ blocks
3	Clue 3: The first chest is buried at the midpoint of the Dock and the Palm. Find its coordinates.	$((-6 + 2) \div 2, (-4 + 2) \div 2) = (-2, -1)$
4	Clue 4: The second chest is at the midpoint of the Palm and the Cave. Find its coordinates.	$((2 + 6) \div 2, (2 + (-4)) \div 2) = (4, -1)$

Part 2: Segment & Angle Clues

#	Task	Answer
5	The Dock, the Rock, and the Cave are on one straight path. DR = 2x + 2 and RC = x + 1, and DC = 12. Find x, DR, and RC.	$3x + 3 = 12 \rightarrow x = 3$; DR = 8, RC = 4
6	At the Palm, ray PD splits ∠LPC. m∠LPD = (3x + 10)°, m∠DPC = (2x + 5)°, and m∠LPC = 95°. Find x and both angle measures.	$5x + 15 = 95 \rightarrow x = 16$; m∠LPD = 58°, m∠DPC = 37°
7	Two paths cross at the Well and form vertical angles of (4y − 20)° and (2y + 30)°. Find y and the angle measure.	$4y - 20 = 2y + 30 \rightarrow y = 25$; 80°
8	What is the measure of an angle that forms a linear pair with the angle from #7?	180° − 80° = 100°

Part 3: Design Your Own Map

#	Task	Answer
9	Place at least four landmarks on a coordinate grid and list their coordinates.	Answers vary.
10	Write a midpoint clue and a distance clue that lead to your treasure. Solve them in an answer key.	Answers vary.
11	Write one angle clue (angle addition or vertical angles) with an answer key.	Answers vary.

Answer Key – CP1

Part 1: Distance & Midpoint Clues

#	Task	Answer
1	Find DP, PC, and DC. Give exact answers and decimals to the nearest hundredth.	DP = 10; PC = $\sqrt{52} = 2\sqrt{13} \approx 7.21$; DC = 12
2	Find the perimeter of the triangle formed by the Dock, the Palm, and the Cave.	$10 + 2\sqrt{13} + 12 = 22 + 2\sqrt{13} \approx 29.21$ blocks
3	The Cave is the midpoint of the Palm and a hidden spot X. Find the coordinates of X.	X = $(2 \cdot 6 - 2, 2 \cdot (-4) - 2) = (10, -10)$
4	Find the midpoint of the Lighthouse and the Cave. Which landmark is closest to it?	$((-6 + 6) \div 2, (4 + (-4)) \div 2) = (0, 0)$. Distances: Palm $\sqrt{8} \approx 2.83$, Rock $\sqrt{20} \approx 4.47$. The Palm is closest.

Part 2: Segment & Angle Clues

#	Task	Answer
5	D, R, and C are collinear with R between D and C. DR = 2x + 2, RC = x + 1, and DC = 12. Find DR and RC. Do these match the map?	x = 3: DR = 8, RC = 4. Yes: the Rock is 8 blocks from the Dock and 4 from the Cave.
6	Ray PD bisects ∠LPC. m∠LPD = (5x − 4)° and m∠DPC = (3x + 12)°. Find m∠LPC.	$5x - 4 = 3x + 12 \rightarrow x = 8$; each part is 36°, so m∠LPC = 72°
7	Two straight paths cross at the Well. One angle is (4y − 20)° and its vertical angle is (2y + 30)°. Find all four angles formed.	y = 25: 80°, 80°, 100°, 100°
8	Two angles at the Rock are complementary. One is 18° more than twice the other. Find both.	$x + (2x + 18) = 90 \rightarrow x = 24$; 24° and 66°

Part 3: Design Your Own Map

#	Task	Answer
9	Place five landmarks and write a clue that uses "the midpoint of A and B is C, find B".	Answers vary.
10	Write a segment addition clue and an angle bisector clue with algebra.	Answers vary.
11	Write the answer key. Trade with a partner and record any clue they found unclear.	Answers vary.

Answer Key – Honors

Part 1: Coordinates & Ratios

#	Task	Answer
1	Find the exact perimeter of triangle DPC and classify the triangle by its sides.	$22 + 2\sqrt{13}$; scalene (10, $2\sqrt{13}$, and 12 are all different)
2	The treasure is ¾ of the way from the Dock to the Palm. Find its coordinates.	$(-6 + \frac{3}{4} \cdot 8, -4 + \frac{3}{4} \cdot 6) = (0, 0.5)$
3	The point Q divides PC so that PQ : QC = 1 : 3. Find Q.	$P + \frac{1}{4}(C - P) = (2 + 1, 2 - 1.5) = (3, 0.5)$
4	The Cave is the midpoint of the Palm and X. Find X, then show that the Rock is not on segment PX.	X = (10, −10). Line PX has slope $(-10 - 2) \div (10 - 2) = -1.5$; at x = 2 it gives y = 2, not −4, so R is not on PX.

Part 2: Equidistant Clues

#	Task	Answer
5	Write the equation of the perpendicular bisector of DP in standard form.	Midpoint (−2, −1); slope of DP = ¾, so the bisector's slope is $-4 \div 3$: $4x + 3y = -11$
6	A clue says the treasure is on the shoreline y = 0 and the same distance from the Dock and the Palm. Find the point two ways: with your bisector…	Set y = 0: $4x = -11 \rightarrow x = -2.75$. Check: $(x + 6)^2 + 16 = (x - 2)^2 + 4 \rightarrow 16x = -44 \rightarrow x = -2.75$. Point (−2.75, 0).
7	Ray PD bisects ∠LPC with $m\angle LPD = (5x - 4)°$ and $m\angle DPC = (3x + 12)°$. Find $m\angle LPC$ and the measure of its supplement.	x = 8: $m\angle LPC = 72°$; supplement 108°
8	Prove that the paths crossing at the Well form two pairs of congruent vertical angles, using the definition of a linear pair.	∠1 and ∠2 form a linear pair, so $m\angle 1 + m\angle 2 = 180°$. ∠2 and ∠3 also form a linear pair, so $m\angle 2 + m\angle 3 = 180°$. So $m\angle 1 = m\angle 3$. The same argument works for the other pair.

Part 3: Design a Unique-Location Map

#	Task	Answer
9	Design a map with five landmarks. Include a ratio (partition) clue and an equidistant clue.	Answers vary.
10	Prove that your clues lead to exactly one point. Explain what would happen if one clue were removed.	Answers vary.
11	Write an answer key and test it with a partner.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original clues correctly use distance, midpoint, and angle relationships and lead to one location. Goes beyond the minimum with a creative, realistic choice.	Original clues correctly use distance, midpoint, and angle relationships and lead to one location.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Map and clue set show labeled coordinates, every clue, and a complete answer key. Polished, organized, and visually engaging.	Map and clue set show labeled coordinates, every clue, and a complete answer key. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.